https://www.technologynetworks.com/applied-sciences/news/breakthrough-in-development-of-drought-resistant-crops-355322

Observing the Effect on Plant Growth of Spinach with the Inoculation of *Priesti megaterium* under Abiotic Stressors

https://www.cropscience.bayer.us/articles/bayer/effects-of-early-season-flooding-in-corn

Introduction

Increased population has led to an increased need for more sustainable and productive methods of obtaining necessary crops.

Current methods using fertilizers and pesticides increase plant growth and yield but reduce sustainability.

Endophytes are bacterial microorganisms that inhabit its host plant and have a mutualistic relationship.

Priestia megaterium is the focus of this project due to its previous research regarding increased plant growth in crops such as soybeans, maize, and apples.

This project aims to fill a gap in research regarding this endophyte by testing its inoculation effects on spinach plants with abiotic stressors such as drought and flooding.

Due to the changing climate, there is an increasing frequency of extreme weather events, so it is crucial to develop resilient crops and agriculture methods to ensure stable food supply.

Objective

Purpose: Observe how inoculation with endophytes change plant tolerance in extreme conditions such as drought and overwatering.

Hypothesis: If *Priestia megaterium* is inoculated into spinach seeds, then the plant growth will increase because the symbiotic relationship between the endophyte and the plant will cause nutrients to be transferred between them and the increased nutrients absorbed by the spinach plant, which will cause an increase in plant growth.

Indepenent Variables:
Endophyte presence in drought conditions
No endophyte presence in drought conditions
Endophyte presence in overwatering conditions
No endophyte presence in overwatering conditions

Dependent Variables:
Visual Stress Score - visual scale (0-3)
Dry Mass g) - Measured at the end of the growing period
Plant Height Percent Change

Constants:
- The amount of light
- The growth system
- The type of plant used
- Time spent in system
- Time spent germinating
- Type of soil

Materials and Methods

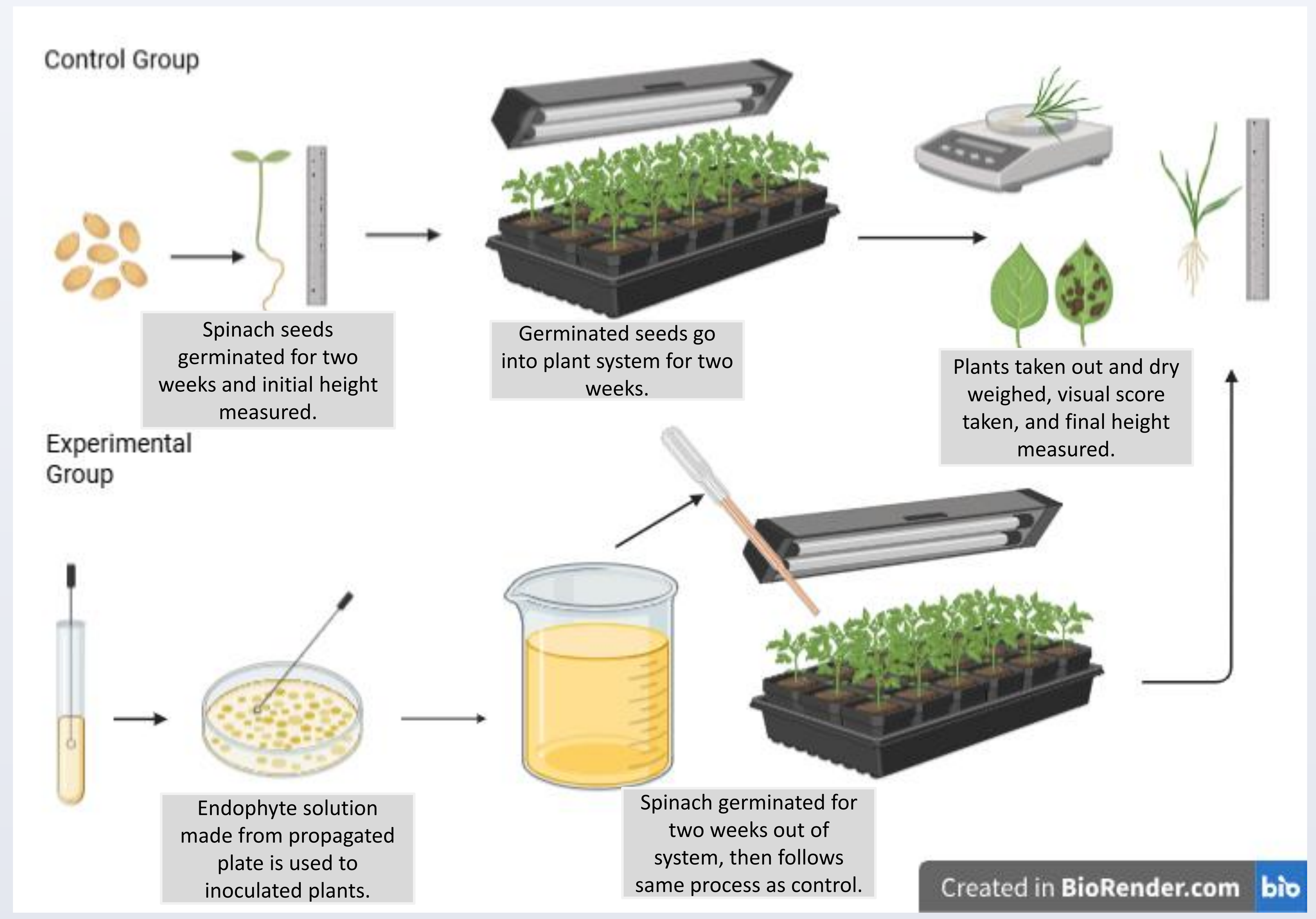
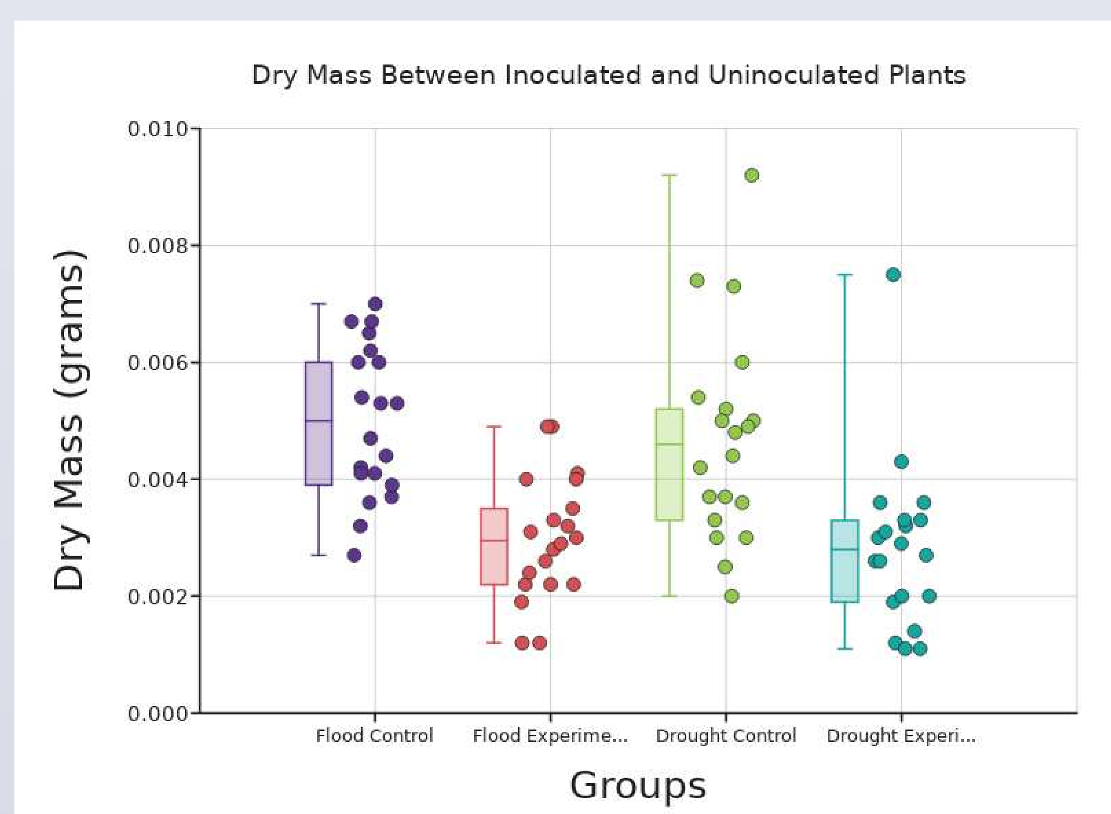
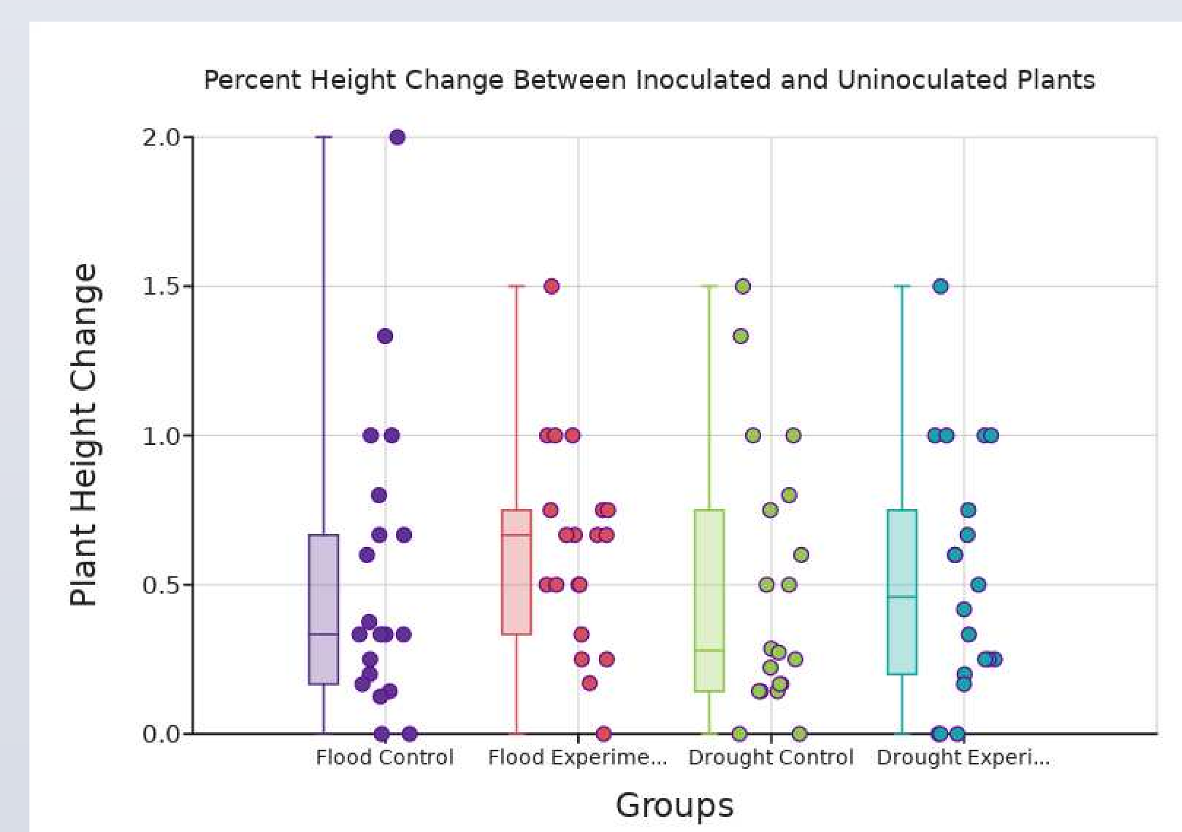


Image taken by researcher showing the plant setup for control group.

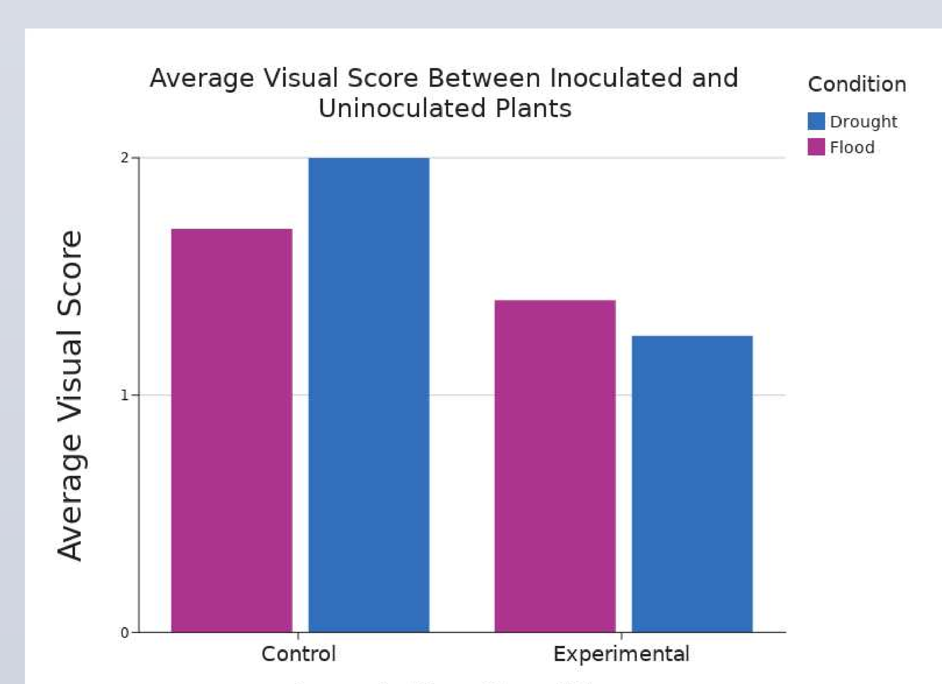
Results



Data visualization showing the difference in dry mass between the experimental groups and control groups.



Data visualization showing the difference in height percent change between experimental and control groups.



Data visualization showing the difference in average visual score between experimental group and control group.

Discussion

- Dry Mass: The ANOVA post-hoc test found that there was a significant difference ($p<0.01$) between the average dry mass of the inoculated and uninoculated plants under the flood and drought abiotic stress conditions.
- Height: The ANOVA post-hoc test found that there was not a significant difference between the average percent height change of the inoculated and uninoculated plants under the flood ($p=0.42$) and drought ($p=0.68$) conditions.
- Visual Score: The ANOVA post-hoc test found that there was a significant difference ($p<0.01$) between the inoculated and uninoculated plants under the drought condition. However, in the flood groups between inoculated and uninoculated groups, there was not a significant difference ($p=0.35$)

Conclusion

The results of this experiment suggest that inoculated spinach plants with *Priestia meaterium* can influence plant growth under abiotic stress conditions. Statistical analysis found a significant difference in dry mass between both conditions and in visual scores under drought conditions.

These findings suggest *that Priestia megaterium* may influence certain aspects of plant growth and appearance during stress conditions, but additional research is needed to determine whether these effects were beneficial and consistent.

One major limitation of this experiment was the relatively short growth period. The spinach plants may not have had enough time to fully develop or to show the long-term effects of inoculation. Future experiment should allow for longer growth period and more trials.

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